

# An Optimum Fast Charging Pattern Search for Li-Ion Batteries Using Particle Swarm Optimization

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**Abstract**—In order to maximize the available performance of the Li-ion batteries, a searching algorithm for an optimal fast charging pattern using particle swarm optimization (PSO) with an evaluation index based on the fuzzy-deduced cost function is proposed in this paper. An optimal five-stage charging strategy is obtained by the PSO searching according to the decision-making in the fitness evaluation index that is computed from the cost function. The cost function formulated by two paramount parameters, charge time and discharge capacity, is employed to assess the cost benefit of the applied charging pattern. Regulating rules of weighting within the cost function are derived from the fuzzy logic inference to attain the best fitness evaluation. The proposed searching methodology for optimal multistage charging pattern features characteristics of fast convergence, effectiveness and easy to implement. Experimental results show that the obtained rapid charging pattern is capable of charging the batteries to 90% available capacity within 51 minutes and also provides 22% more cycle life than the conventional constant current–constant voltage (CC-CV) method.

**Keywords**—five-step charge; fuzzy logic control; Li-ion battery; particle swarm optimization

## I. INTRODUCTION

Rechargeable batteries play a significant part in energy storage solutions for the state-of-the-art electronic technologies. Among the commonly used rechargeable batteries (lead-acid, nickel-based, and lithium-ion chemistries), the lithium-ion (Li-ion) batteries are the most popular choice for applications in modern electronic appliances such as portable consumer electronics, aerospace, transportation vehicles, and energy backup for renewable power systems etc., by reason of their attractive characteristics like environmentally friendly benefit, high energy and power density, high open-circuit voltage (OCV), low self-discharge rate, and immunization of memory effect [1]–[8].

The charging profile of the Li-ion batteries using the CC-CV charging method, as shown in Fig. 1, can be partitioned into four operating regions: 1) trickle charge, 2) constant current, 3) constant voltage, and 4) end of charge. Trickle charging is worked only if the battery is discharged deeply, i.e. battery voltage is less than 3 V. Above 3.0 V, the battery is charged at a constant current rate until the cell voltage rises to a default value (4.1 or 4.2 V). When the default voltage has been approached, the charging profile turns into the constant-voltage phase. In this region, the charging current should be gradually decreased as the battery voltage approaches 4.2 V. Charging current should be progressively decreased until a certain threshold is met—typically about 2% of the rated battery

capacity [1]. As soon as the charging current is reached, the charger enters the end-of-charge region. The third phase of the cycle completes the normal charging of the battery at a tardily reduced rate; as a result, it significantly prolongs the charging time and also reduces the cycle life of the battery. To deal with this problem, several efforts have been devoted to the study of the charging techniques and the charging strategies [9]–[11]. These methods can achieve better charging efficiencies and shorter charging time. However, these techniques are sophisticated and require digital controllers for realization. Therefore, they are unsuitable to be integrated into commercial battery charging ICs. In addition, battery cycle life issues are seldom discussed in these researches. The multistage constant-current charging method with intelligent charge strategy has been confirmed to possess the advantages such as longer cycle life, higher charge/discharge energy efficiency, and shorter charging time [12], [13]. Moreover, these methods are easy to embed into the power management IC because of its simplicity.

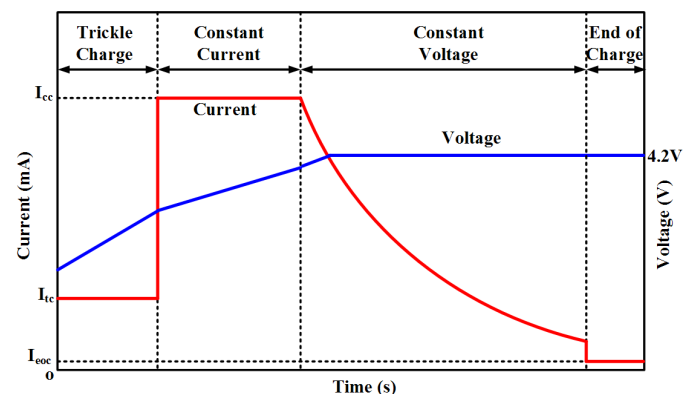


Figure 1. Typical charging profile of a Li-ion battery using CC-CV method.

Searching for the optimal charging pattern can be regarded as combinatorial optimization problems. One possible method to attain the optimal charging pattern is to try every combination of the charging current value. Hence, the evolutionary computation techniques such as ant colony system (ACS), genetic algorithm (GA), and particle swarm optimization (PSO), can be employed to resolve this kind of problems. Among these solutions, the PSO is developed by Eberhart and Kennedy in 1995, which was inspired by social behavior of bird flocks and fish schools. The PSO is a population-based evolutionary computational technique, which features several attractive merits over other optimization techniques, for example: 1) Unlike many other meta-heuristic techniques, the PSO has fewer parameters to tune and adjust; 2) the PSO features the characteristics of fast convergence and

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less computational time; 3) the PSO is a non-gradient, derivative-free algorithm, such that the objective function of the problem to be optimized is not necessarily continuous, convex or differentiable; 4) the algorithm can estimate several solutions in a single iteration, which minimizes the probability of the search getting trapped in local minimum; 5) unlike deterministic techniques, the solution obtained by the PSO is not related to the initial solution; and 6) the PSO algorithm is simple to comprehend, easy to implement, and flexible to program since it employs basic mathematical and Boolean logic operations. On the other hand, the PSO has demerits such as it lacks rigid mathematical foundation, and as most meta-heuristic methods are, it is a highly problem-dependent method. In order to guarantee a good quality solution for every system, the PSO parameters need to be regulated and updated. As a result, how to select good parameters (for example, the number of particles and the learning factors) for the PSO algorithm still presents a challenge for inexperienced users.

A searching algorithm for an optimal five-stage fast charging pattern using PSO with a fuzzy-deduced cost function evaluation index is proposed in this paper to maximize the available performance of the Li-ion batteries. According to the decision-making, the evaluation index that is derived from the cost function directs the PSO to find out the optimal solution. The cost function formulated by two paramount parameters, i.e. charge time and discharge capacity, is employed to assess the cost effectiveness of the prospective pattern within the solution space. Regulating rules of weighting within the cost function are derived from the fuzzy logic inference to attain the best evaluation. **The proposed searching method** features characteristics of fast convergence, effectiveness, and easy to implement. The objective function for problem formulation, design considerations, parameter setting procedure, and experimental setup and results are described in detail in the following Sections.

## II. BASIC PHILOSOPHY OF PSO ALGORITHM

The PSO is a symbiotic cooperation algorithm, which was evolved into a stochastic optimization search process, based on modified social behavior simulation models of the bird flocks. Hence, it can serve as a powerful optimizer to deal with global optimization problems having single- or multi-objective functions with several local optimum values. Prior researches prove that the PSO technique has been successfully applied for solving science or technology optimization problems and have been shown its promising potential. The most remarkable property of the PSO approach is its simplicity because only two model equations are involved. In order to improve the poor convergence characteristic, various evolutions of the PSO method are presented, yet the most popular is the global formulation ( $G_{best}$  model) devised by Shi and Eberhart.

Ingredients dominating the flying experience of each particle in its search for optimal solution are shown in Fig. 2. Many cooperative particles are distributed inside the search space; each particle in the coordinate denotes a probable solution relative to two vectors, the velocity ( $V_i$ ) and position ( $X_i$ ) vectors. In a  $d$ -dimensional search space,  $X_i = [x_{i1}, x_{i2}, \dots, x_{id}]$  and  $V_i = [v_{i1}, v_{i2}, \dots, v_{id}]$  are the two vectors associated with each particle  $i$ . A swarm is composed of several of particles or

“solution candidates” which fly through the feasible solution space to search locations in which optimal solutions exist. Every particle will share or exchange information with the rest of the swarm in a certain communication way during their exploration. The interaction aims at optimizing their search experience, the experience contains individual cognitive share and global social share that are acquired in each individual search process. For each iteration, the particle which has the best solution shares its position coordinates ( $G_{best}$ ) message with the rest of the swarm members. Then, each particle updates its flying direction based on its own best search experience ( $P_{best}$ ) and ( $G_{best}$ ) according to the model equations as follows:

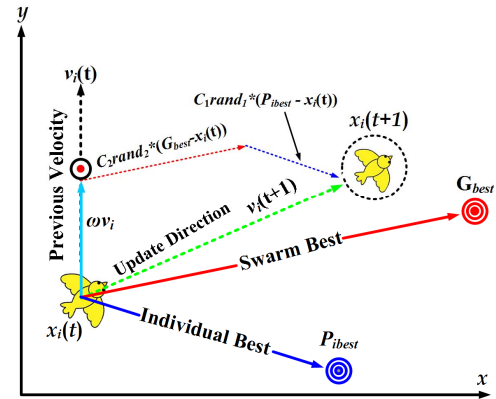


Figure 2. Concept of update in a searching particle by PSO.

$$v_{id}^{k+1} = \underbrace{w v_{id}^k}_{\text{previous velocity}} + \underbrace{C_1 rand_1^k (P_{best_{id}}^k - x_{id}^k)}_{\text{cognitive component}} + \underbrace{C_2 rand_2^k (G_{best_d}^k - x_{id}^k)}_{\text{social component}} \quad (1)$$

$$x_{id}^{k+1} = x_{id}^k + v_{id}^{k+1} \quad (2)$$

Where  $v_{id}$  is the value of dimension  $d$  in the velocity vector for particle  $i$ .  $x_{id}$  is the position of particle  $i$  along dimension  $d$ .  $w$  is the selected inertia weight. The acceleration constants  $C_1$  and  $C_2$  are the cognitive and social learning rates, respectively, which change the velocity of a particle towards  $P_{best}$  and  $G_{best}$ . When they are set to be equal, it keeps balance between the particle's individual and social behavior. The  $rand_1$  and  $rand_2$  are random values on the range  $[0, 1]$  that are added in the model to introduce stochastic nature to the particles' movement.  $G_{best}^k$  is the current location along dimension  $d$  of the neighborhood particle with the best fitness; i.e.,  $G_{best}^k = [x_1^{G_{best}}, x_2^{G_{best}}, \dots, x_d^{G_{best}}]$ .  $P_{best}_i^k$  is the location along dimension  $d$  at which the particle previously had the best fitness measure;  $P_{best}_i^k = [x_{i1}^{P_{best}}, x_{i2}^{P_{best}}, \dots, x_{id}^{P_{best}}]$ , and  $k$  is the iteration index.

As a result, the next position of particle  $i$  is updated by (2) based on the velocity modification in (1) consists of three terms. 1) Previous velocity: changing velocity values will change the particle direction, causing the particle to make a turn between successive iterations. The velocity aims at controlling the search resolution and fitness. If velocity is too high, then particles will move beyond a good solution, and if velocity is too low, particles will be trapped in local minima; 2) Cognition

component: this term stands for the self-cognition ability of each particle to make the best position search achievable based on its own flying experience; 3) Social component: this component coincides with each particle tendency to be drawn toward the best solution found among the whole individuals inside a population.

### III. PSO-BASED FAST CHARGING PATTERN SEARCH

#### A. Five-Stage Constant-Current Charging Mechanism

The conventional CC-CV charge is not suitable for rapid charging due to the constant voltage charging phase severely extends the charging time and also reduces the cycle life of the battery. The features of the short charging time, extendable cycling life and efficient energy transfer are the major measure of a high-quality rapid charging algorithm. The multistage charging technique is proven to have the salient performance to satisfy the fast-charge criteria in the early works. Therefore, in this paper, the five-stage charging approach is utilized to do rapid charge mechanism for the Li-ion batteries. Shown in Fig. 3 is the conception of the multistage constant current charging algorithm. From Fig. 3, the total charging period is divided into five stages. In each stage, the charging current is set to a preset value. The battery voltage will increase gradually during the charging phase. When the voltage reaches the default limit voltage ( $V_{lim}$ ), the charge process will enter the next phase and a new charging current level will be applied accordingly. This process will continue until the charging phase 5 is completed.

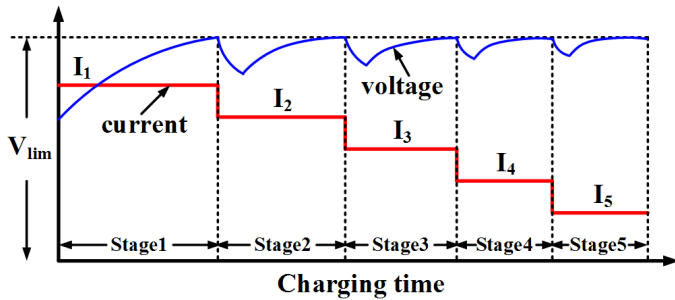


Figure 3. Conception of multistage constant-current charging algorithm.

#### B. Problem Formulation

Generally, an accurate behavior of the Li-ion batteries is not easy to model mathematically due to their complicated electrochemical characteristics. Accordingly, it is also hard to determine the optimal current level for each stage only by battery models. In addition, the battery charging time (CT) and the discharge capacity percentage (DCP) are mainly dependent on the level of the charging C rate. In which the DCP is defined as the percentage ratio of the discharge capacity obtained by the five-stage charge to that by the CC-CV method, both with the 0.1 C discharge rate; and the C represents the battery capacity expressed in terms of ampere-hours (Ah). The maximum charging current for most commercially available Li-ion batteries are lower than 2.5 C. Moreover, the fast charging criteria require that, under the safe condition, the fully charged battery must have the maximum discharge capacity within the shortest charging time.

In order to determine the optimal five-stage charging C rates to meet the rapid-charge criteria and achieve the best charging cost benefit, an optimum charge pattern search strategy using PSO algorithm is proposed in this paper. The formulation of this problem can be described as

$$\text{Maximize } f_{\text{cost}}^i(T_{\text{ch}}^i, C_{\text{dis}}^i) = \alpha T_{\text{ch}}^i(\bar{I}_j^i) + \beta C_{\text{dis}}^i(\bar{I}_j^i) \quad \bar{I}_j^i \in \mathbb{S}$$

$$\text{where } \bar{I}_j^i = [I_1^i, I_2^i, I_3^i, I_4^i, I_5^i]$$

$$\text{Subject to } 30 \leq T_{\text{ch}}^i(\bar{I}_j^i) \leq 90 \text{ min} \quad (3)$$

$$80\% \leq C_{\text{dis}}^i(\bar{I}_j^i) \leq 100\%$$

$$I_j^i \geq I_k^i \quad \text{if } j < k, \quad j, k = 1, 2, \dots, 5$$

Where the cost function  $f_{\text{cost}}^i$  for battery  $i$  is function of charging time ( $T_{\text{ch}}^i$ ) and discharge capacity percentage ( $C_{\text{dis}}^i$ ), and both the charging time and discharge capacity percentage are related to the five-stage charging current value  $\bar{I}_j^i$ .

$\mathbb{S}$  represents the set of all feasible charging patterns.  $\alpha, \beta$  are two weightings, they are used to highlight the impact of the charging time and discharge capacity percentage on the cost function. Optimal regulating rules of the weightings are carried out by the fuzzy logic control in this paper to attain the best fitness evaluation. Subsection C will depict the control mechanism of how to allocate the weighting factors to complete the cost function optimization.

Therefore, the objective function of the battery charging problem studied in this paper is formulated by maximizing the cost-effective function that is composed of the charge time and discharge capacity percentage. Any constraints associated with the optimization problem could be incorporated into the objective function as penalty functions. That is, the target charging time and discharge capacity percentage are selected to the range [30, 90] min and [80, 100] %, respectively, to meet the common requirements of the fast charging standard. On the other hand, it should be noted that the current level of the present stage should not surpass the current setting of the previous stage. Because the voltage-limit value of each stage is the same, if the charging current of the present stage is greater than that of the previous stage, the battery voltage will reach the voltage limit instantly; thus, the duration time of this stage will be very short.

#### C. Fitness Value Evaluation with Fuzzy Logic Control

When each battery (particle) has completed its five-stage charge process for each iteration, the fitness value of each battery should be calculated. Fitness value of the optimization problem is used to evaluate the charging cost benefit of the applied charge pattern for each battery inside the swarm to judge whether the trial charge pattern is the best or not. Comparing each battery's fitness with its previous best fitness ( $P_{\text{best}}$ ) obtained. If the present value is better than old  $P_{\text{best}}$ , then update  $P_{\text{best}}$  and the location data with the current fitness value and location in the  $d$ -dimensional space. The next compares  $P_{\text{best}}$  of all particles with each other and update the swarm global best location with the greatest fitness ( $G_{\text{best}}$ ). As mentioned in the previous subsection, the objective of this problem in this paper is to maximize the charging cost function,



i.e. the current pattern make the battery delivers the most discharge capacity within the shortest charging time, and then this charging pattern will obtain the maximum cost benefit. Hence, the fitness function  $F^i(\bar{I}_j^i)$  for the battery  $i$  can be expressed as

$$F^i(\bar{I}_j^i) = \alpha T_{ch}^i(\bar{I}_j^i) + \beta C_{dis}^i(\bar{I}_j^i) \quad (4)$$

However, according to the understanding of the battery's electrochemical characteristics and the field-tested experience, we know that the charging time and discharge capacity are strongly depend on the charging/discharging current levels and they are conflictive each other nearly. Therefore, how to correctly select the weighting factors  $\alpha$  and  $\beta$  in (4) to obtain the maximum charging cost effectiveness is significantly critical here. As mentioned above, the fifty-percent ratio for  $\alpha$  and  $\beta$  is not the best solution obviously. In this paper, optimal regulating rules of weighting factors are formulated by the fuzzy logic (FL) inference to carry out the best fitness value evaluation. Shown in Fig. 4 is the scheme of the proposed fuzzy controller for the fitness value assessment. Form Fig. 4, the five-stage charging time and discharge capacity percentage are used as the input parameter, and the output parameter is evaluative value of the fitness (cost) function. The Mamdani-type minimum inferential method cooperating with the center-of-area (COA) defuzzification procedure is utilized to produce the crisp controller output. The membership functions (MFs) corresponding to the charging time, discharge capacity percentage, and fitness evaluation value are illustrated in Fig. 5(a)-(c), respectively. Here in all three MFs, except for the two fuzzy sets at the outmost ends (trapezoidal MFs are considered), symmetric triangles with equal base and fifty-percent overlap with adjacent MFs are chosen. The universe of discourse (UOD) in the MFs of the input variables,  $T_{ch}^i$  and  $C_{dis}^i$ , are defined on the domain [30, 90] min. and [80%, 100%], respectively, to agree with the constraints specified in the problem formulation. In addition, the UOD in the MF of the output variable,  $\Delta F_N^i$ , is defined on the normalized domain [0, 1].

The linguistic value S denotes short or small in charging time or discharge capacity and L represents long or large in charging time or discharge capacity respectively. Such that the SSS, SS, S, MS, M, ML, L, LL, and LLL stand for tiny, very small, small, medium small, medium, medium large, large, very large, and huge, respectively. The FL controller structure approximates the way of expert operation intuitiveness; this makes it attractive and easy to incorporate heuristic rules that reflect the experience of human experts into the controller. The rule bases for deducing the fitness value is shown in Table I. This is a widely used rule base designed with a two-dimensional phase plane. As shown in Table I, the control rules are reasoned to obtain the optimal cost function based on the perception that the larger discharge capacity and the shorter charging time will obtain the higher cost benefit, and on the contrary, the smaller discharge capacity and the longer charging time will result in the lower cost benefit. For example, if the discharge capacity is large and the charging time is short,

then the fitness evaluation is huge. The rule can be formed as: If  $C_{dis}^i$  is L and  $T_{ch}^i$  is S, then  $F^i(\bar{I}_j^i)$  is LLL.

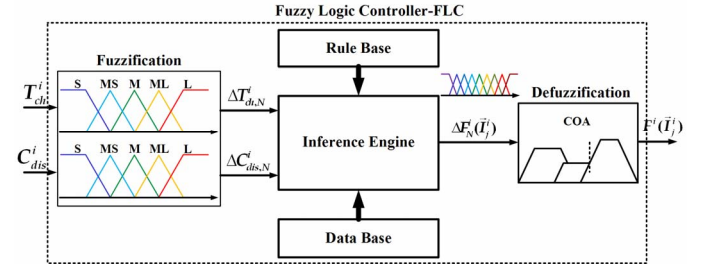


Figure 4. Scheme of the proposed fuzzy controller.

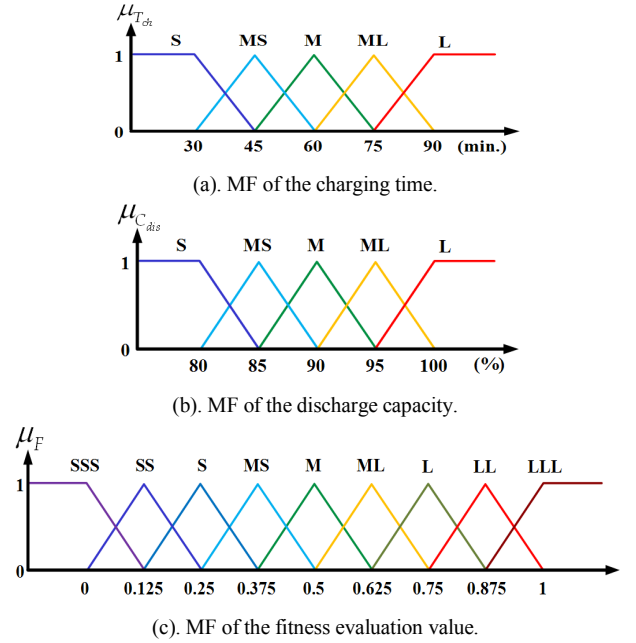


Figure 5. MFs of the proposed fuzzy controller.

TABLE I. RULE BASE FOR DERIVING FITNESS VALUE

| $T_{ch}^i \backslash C_{dis}^i$ | S             | MS           | M            | ML           | L            |
|---------------------------------|---------------|--------------|--------------|--------------|--------------|
| S                               | M<br>Rule1    | ML<br>Rule2  | L<br>Rule3   | LL<br>Rule4  | LLL<br>Rule5 |
| MS                              | MS<br>Rule6   | M<br>Rule7   | ML<br>Rule8  | L<br>Rule9   | LL<br>Rule10 |
| M                               | S<br>Rule11   | MS<br>Rule12 | M<br>Rule13  | ML<br>Rule14 | L<br>Rule15  |
| ML                              | SS<br>Rule16  | S<br>Rule17  | MS<br>Rule18 | M<br>Rule19  | ML<br>Rule20 |
| L                               | SSS<br>Rule21 | SS<br>Rule22 | S<br>Rule23  | MS<br>Rule24 | M<br>Rule25  |

#### D. PSO-Based Search Procedure

In the PSO algorithm, the potential solution acts as a particle, and it imitates how natural creatures behave as a swarm. Each particle has a fitness value, which is evaluated by the fitness function to be optimized, and a velocity directs the trajectory of the particle. By this way, the individual particles are ultimately attracted toward their previous best location and the swarm's best solution in searching space.

Shown in Fig. 6 is the operating flowchart of the PSO algorithm applied to the optimal fast charging pattern search

for Li-ion batteries in this paper. The main operating procedures are described as follows:

- 1) Battery screening test: All batteries had to pass screening tests with acceptable open-circuit voltage (OCV), impedance, weight, and rated capacity values.
- 2) System initialized and PSO parameter design: Initialize the searching space according to the number of stages and charging current candidates and choose the parameters for PSO. In this paper, the initial swarm of batteries are first randomly generated. A random number generator is programmed to generate the initial five-stage charging current patterns of the batteries within the feasible search space. The charging pattern of each battery is also outputted to a file for subsequent use.
- 3) Charging/discharge batteries: The charging process was carried out by means of a WonATech WBCS3000 programmable charge/discharge tester. When the charging process is completed with the applied charging patterns, the discharging processes are also conducted using the same equipment. The obtained discharge capacity and charging time for each battery are recorded in an output file for fitness evaluation.
- 4) Fitness evaluation: According to the obtained discharge capacity and charging time, based on the fuzzy logic inference, to compute the fitness value (cost function) of each battery in the swarm.
- 5)  $P_{best}$  and  $G_{best}$  update: For each time search, comparing each battery's fitness with its previous best fitness ( $P_{best}$ ) obtained. If the current value is better than old  $P_{best}$ , then update  $P_{best}$  and the location data with the current fitness value and location. The next compares  $P_{best}$  of all batteries with each other and update the swarm global best location with the greatest fitness ( $G_{best}$ ).
- 6) Generate new charging Pattern: If the termination condition is unreachable, the incremental value of each stage charging current (velocity) for each battery will be changed towards its  $P_{best}$  and  $G_{best}$ . This acceleration is weighted by a random term. Then, a new charging pattern in the solution space is generated for each battery by adding the new incremental current value to the previous charging current of each battery according to (1) and (2).
- 7) Repeat steps (3) to (5) until convergence is reached based on the two criteria: 1) the difference of the average fitness value between the before and after iteration is less than 0.005; 2) the standard deviation (STDEV) less than 0.01.

#### IV. EXPERIMENTAL SETUP AND RESULTS

Shown in Fig. 7 is the configuration of the experimental system. The charging/discharging equipment used in this paper is WBCS3000 (5A, 5V, 8 channel) from WonATech corporation. Sixteen commercially available Li-ion batteries for notebooks are utilized in these experiments. All the batteries tested in this paper have similar electrochemical characteristics and passed screening tests. The right-hand side of the dotted line is the software part, where the PSO algorithm is implemented. The left-hand side of the dotted line is the hardware part, which primarily is a battery tester with

programmable ability. The relative parameters in the PSO method are: the  $w$  is 0.1;  $C_1$  and  $C_2$  are equal to 2;  $rand_1$  and  $rand_2$  are random numbers between 0 and 1. The specification of the test battery is 3.6V, 2200 mAh.

Shown in Table II is first test result. The fitness value (fuzzy output),  $P_{best}$ ,  $G_{best}$ , and the five-stage current setting values are tabulated. According to the (1) and (2), the incremental current value and new charging pattern are generated. This process is continuous until the convergence criteria are met. At the seventh test, the difference (0.005) of the average fitness value between the sixth and seventh iteration is less than 0.05; and the standard deviation (0.007089) (STDEV) less than 0.01, as shown in Table III. Therefore, stop the searching procedure. Shown in Fig. 8-10 are the convergence curves corresponding to the cost function, charging time, and discharge capacity, respectively. Obviously, the current setting values in battery number 7 are the optimal fast-charge pattern.

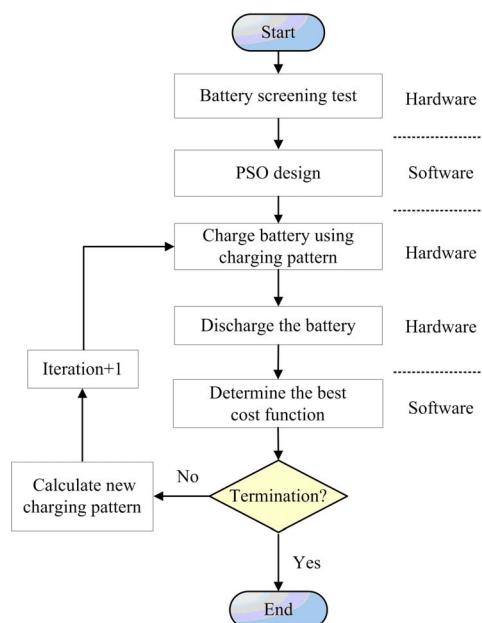


Figure 6. Flowchart of the proposed method.

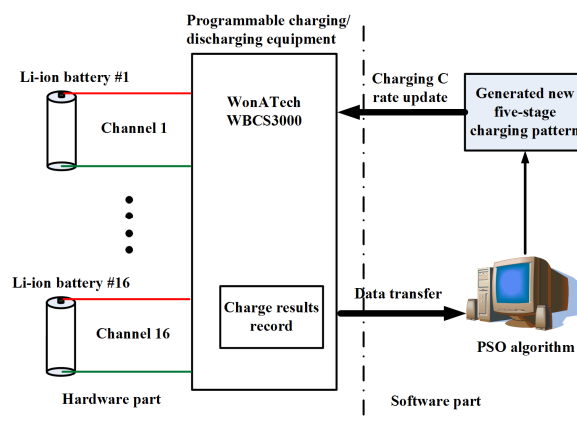


Figure 7. Experimental system configuration.

TABLE II. FIRST TEST RESULT

| Battery no. | Fuzzy Output1 | Pbest | Gbest | Battery no. | I <sub>1</sub> | I <sub>2</sub> | I <sub>3</sub> | I <sub>4</sub> | I <sub>5</sub> |
|-------------|---------------|-------|-------|-------------|----------------|----------------|----------------|----------------|----------------|
| B1          | 0.516         |       |       | B1          | 1.32           | 1.23           | 0.96           | 0.52           | 0.43           |
| B2          | 0.517         |       |       | B2          | 1.41           | 0.97           | 0.72           | 0.68           | 0.24           |
| B3          | 0.417         | 0.417 |       | B3          | 0.89           | 0.74           | 0.53           | 0.41           | 0.12           |
| B4          | 0.502         | 0.502 |       | B4          | 1.08           | 1.02           | 0.53           | 0.49           | 0.37           |
| B5          | 0.515         | 0.515 |       | B5          | 1.17           | 1.02           | 0.91           | 0.56           | 0.45           |
| B6          | 0.556         | 0.556 |       | B6          | 1.22           | 1.19           | 0.65           | 0.41           | 0.31           |
| B7          | 0.596         | 0.596 | 0.596 | B7          | 1.44           | 1.18           | 0.87           | 0.71           | 0.41           |
| B8          | 0.472         | 0.472 |       | B8          | 1.49           | 0.96           | 0.81           | 0.34           | 0.12           |
| B9          | 0.404         | 0.404 |       | B9          | 1.36           | 1.20           | 1.15           | 0.40           | 0.12           |
| B10         | 0.561         | 0.561 |       | B10         | 1.39           | 1.16           | 0.66           | 0.54           | 0.31           |
| B11         | 0.488         | 0.488 |       | B11         | 1.36           | 1.25           | 1.14           | 0.85           | 0.25           |
| B12         | 0.479         | 0.479 |       | B12         | 1.43           | 0.83           | 0.70           | 0.26           | 0.15           |
| B13         | 0.501         | 0.501 |       | B13         | 1.27           | 1.22           | 0.60           | 0.53           | 0.50           |
| B14         | 0.474         | 0.474 |       | B14         | 1.05           | 0.71           | 0.35           | 0.24           | 0.21           |
| B15         | 0.432         | 0.432 |       | B15         | 0.98           | 0.97           | 0.89           | 0.49           | 0.35           |
| B16         | 0.515         |       |       | B16         | 1.40           | 1.25           | 0.74           | 0.66           | 0.50           |
| MAX         | 0.596         | 0.596 |       |             |                |                |                |                |                |
| MIN         | 0.404         |       |       |             |                |                |                |                |                |
| STDEV       | 0.051419      |       |       |             |                |                |                |                |                |
| AVG         | 0.496563      |       |       |             |                |                |                |                |                |
|             |               |       |       | B7          | 1.44           | 1.18           | 0.87           | 0.71           | 0.41           |

TABLE III. RESULTS FORM THE FIRST TO THE SEVENTH TEST

| Battery no. | Fuzzy Output1 | Fuzzy Output2 | Fuzzy Output3 | Fuzzy Output4 | Fuzzy Output5 | Fuzzy Output6 | Fuzzy Output7 | Pbest | Gbest | Battery no. | I <sub>1</sub> | I <sub>2</sub> | I <sub>3</sub> | I <sub>4</sub> | I <sub>5</sub> |
|-------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|-------|-------|-------------|----------------|----------------|----------------|----------------|----------------|
| B1          | 0.576         | 0.586         | 0.559         | 0.537         | 0.538         | 0.551         | 0.516         | 0.586 |       | B1          | 1.43           | 1.12           | 0.87           | 0.73           | 0.39           |
| B2          | 0.581         | 0.566         | 0.558         | 0.543         | 0.531         | 0.557         | 0.517         | 0.581 |       | B2          | 1.43           | 1.18           | 0.86           | 0.71           | 0.40           |
| B3          | 0.593         | 0.567         | 0.584         | 0.546         | 0.545         | 0.572         | 0.417         | 0.593 |       | B3          | 1.44           | 1.33           | 0.88           | 0.69           | 0.38           |
| B4          | 0.583         | 0.597         | 0.557         | 0.553         | 0.561         | 0.548         | 0.502         | 0.597 |       | B4          | 1.50           | 1.16           | 0.86           | 0.74           | 0.41           |
| B5          | 0.582         | 0.591         | 0.592         | 0.554         | 0.559         | 0.501         | 0.515         | 0.592 |       | B5          | 1.47           | 1.17           | 0.88           | 0.72           | 0.41           |
| B6          | 0.577         | 0.566         | 0.543         | 0.541         | 0.546         | 0.557         | 0.556         | 0.577 |       | B6          | 1.45           | 1.19           | 0.79           | 0.71           | 0.40           |
| B7          | 0.594         | 0.598         | 0.592         | 0.564         | 0.569         | 0.577         | 0.596         | 0.598 | 0.598 | B7          | 1.44           | 1.18           | 0.87           | 0.71           | 0.41           |
| B8          | 0.586         | 0.550         | 0.556         | 0.543         | 0.542         | 0.589         | 0.472         | 0.589 |       | B8          | 1.50           | 1.03           | 0.99           | 0.62           | 0.31           |
| B9          | 0.581         | 0.572         | 0.593         | 0.546         | 0.553         | 0.557         | 0.404         | 0.593 |       | B9          | 1.49           | 1.21           | 0.79           | 0.68           | 0.39           |
| B10         | 0.581         | 0.588         | 0.541         | 0.530         | 0.519         | 0.553         | 0.561         | 0.588 |       | B10         | 1.50           | 1.03           | 0.71           | 0.52           | 0.40           |
| B11         | 0.580         | 0.592         | 0.593         | 0.531         | 0.546         | 0.488         | 0.488         | 0.593 |       | B11         | 1.46           | 1.13           | 1.10           | 0.66           | 0.38           |
| B12         | 0.572         | 0.563         | 0.558         | 0.526         | 0.529         | 0.514         | 0.479         | 0.572 |       | B12         | 1.44           | 1.20           | 0.87           | 0.77           | 0.41           |
| B13         | 0.585         | 0.586         | 0.579         | 0.541         | 0.537         | 0.532         | 0.501         | 0.586 |       | B13         | 1.45           | 1.19           | 0.88           | 0.71           | 0.41           |
| B14         | 0.575         | 0.581         | 0.570         | 0.515         | 0.531         | 0.529         | 0.474         | 0.581 |       | B14         | 1.49           | 1.17           | 0.92           | 0.70           | 0.41           |
| B15         | 0.570         | 0.553         | 0.550         | 0.516         | 0.517         | 0.497         | 0.432         | 0.570 |       | B15         | 1.45           | 1.19           | 0.87           | 0.71           | 0.41           |
| B16         | 0.570         | 0.548         | 0.559         | 0.511         | 0.520         | 0.519         | 0.515         | 0.570 |       | B16         | 1.46           | 1.17           | 0.88           | 0.65           | 0.41           |
| MAX         | 0.594         | 0.598         | 0.593         | 0.564         | 0.569         | 0.589         | 0.596         | 0.598 |       |             |                |                |                |                |                |
| MIN         | 0.570         | 0.548         | 0.541         | 0.511         | 0.517         | 0.497         | 0.404         |       |       |             |                |                |                |                |                |
| STDEV       | 0.00709       | 0.016775      | 0.01842       | 0.01493       | 0.015739      | 0.024481      | 0.051419      |       |       |             |                |                |                |                |                |
| AVG         | 0.58038       | 0.575250      | 0.567750      | 0.53731       | 0.539563      | 0.543813      | 0.496563      |       |       | B7          | 1.44           | 1.18           | 0.87           | 0.71           | 0.41           |

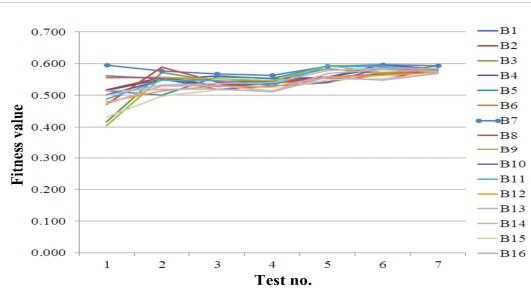


Figure 8. Convergence curve of fitness value.

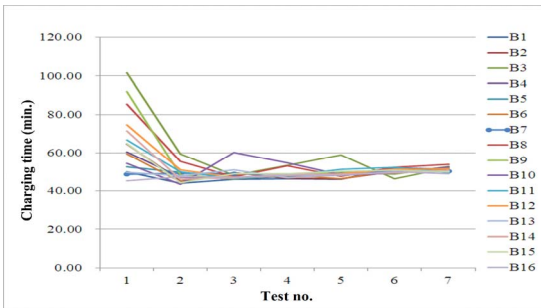


Figure 9. Convergence curve of charging time.

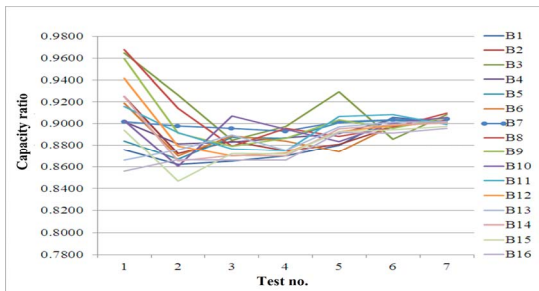


Figure 10. Convergence curve of discharge capacity ratio.

## V. CONCLUSIONS

Li-ion batteries are damageable by reason of the overvoltage and improper charge C rate. Efficacies of the charging strategy and electrified pattern dominate their available performance and lifespan intensely. Superior charging strategies and current patterns make the desired properties in high-quality chargers such as longer cycle life, higher charge/discharge capacity, and shorter electrified time realizable. In order to maximize the available performance of the Li-ion batteries, a searching algorithm for an optimal fast charging pattern using particle swarm optimization (PSO) with a fuzzy-deduced cost function evaluation index is proposed in this paper. Experimental results show that the obtained rapid charging pattern is capable of charging the batteries to 90% available capacity within 51 minutes and also provides 22% more cycle life than the conventional CC-CV method.

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